

first welfare theorem

private market equilibrium is pareto efficient if no market failures

impossible to find allocation that improves everyone's welfare

Public Economics (ECON 131)

Section #4: Efficiency Costs of Taxation

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second welfare theorem

any pareto efficient allocation can be achieved by lump-sum taxes + transfers based on individual ability (potential)

↳ in practice, difficult to do and will distort the market equilibrium

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1 Taxation and Efficiency

The Second Welfare Theorem tells us that the ideal way to tax somebody (moral perceptions of fairness aside) is based on their innate characteristics—i.e. things that are impossible for them to change with their behavior—but, this is easier said than done. Suppose we tried to enact an income tax based on people's potential earnings. The government would somehow have to know how skilled the person is at every job, measuring things from intelligence to health to interpersonal skills, and to figure out how these things would translate into the labor market.

potential vs actual earnings

Instead of doing all that, the government imposes taxes based on outcomes, that is, how much money you actually make, what you purchase and how well your investments do. When we tax things that people can control, we risk distorting incentives in that market. If we tax wages, for instance, people might choose to work more or less than is optimal. People can go to great lengths to avoid taxes. One dramatic instance of tax avoidance with negative consequences is documented in Oates and Schwab (2015) and retold in the Gruber textbook. In 1696, King William III of England instituted a tax based on the number of windows in a house. In response, people boarded up and closed off windows, reducing ventilation, and aiding the spread of disease. Other, more modern examples of tax avoidance include ABBA dressing oddly to write-off their clothing purchases and White Claw being fermented like a beer (in part) to avoid the higher taxes associated with vodka-sodas.

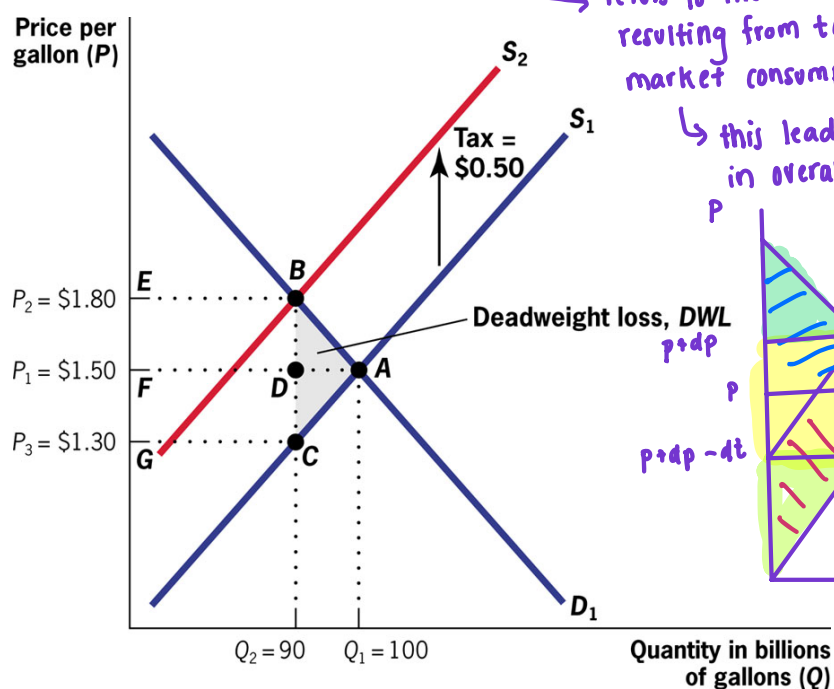
this type of tax leads to distortions (ppt & behavior), which leads to inefficiency

Of course, taxation can also be used to fix distortions that already exist in the market. For instance, many economists and environmentalists advocate for a carbon tax. In this case, the price of carbon-intensive goods and services would increase to better reflect the externalities these products have on the health of the planet and we who inhabit it. In other words, taxes could make the market function *more* efficiently.

→ is a result of inefficiencies

Deadweight burden (DWB, also called excess burden or deadweight loss (DWL)) of taxation is defined as the welfare loss (measured in dollars) created by a tax, relative to the baseline welfare in the free market.

In a simple supply and demand diagram, welfare is measured by the sum of the consumer surplus and producer surplus. We can measure the DWL of taxation as the change in consumer + producer surplus minus tax collected: it is the triangle on the figure, which economists call the **Harberger Triangle**:



refers to the economic inefficiency resulting from taxes that distort market consumption/production

→ this leads to a reduction in overall economic welfare

Mathematically, area of this triangle is given by:

check math below if you want

$$DWB = \frac{1}{2} dQ \cdot dt = \frac{1}{2} \cdot \frac{\epsilon_S \cdot \epsilon_D}{\epsilon_S - \epsilon_D} \cdot \frac{Q}{p} (dt)^2$$

Looking at the formula, we can see that if there is **no change in quantities consumed**, the tax has **no efficiency costs**. The inefficiency of any tax is determined by the extent to which consumers and producers change their behavior to avoid the tax. Deadweight loss is caused by individuals and firms making **inefficient consumption and production choices** in order to avoid taxation.

→ $(dt)^2$ so if $(2dt)^2 \Rightarrow 4dt^2$

We can also observe that the deadweight loss increases with the square of the tax, meaning that a **tax twice as big will result in a deadweight loss that is four times as big**.

if $dQ = 0$, $DWL = ?$ [0]

perfectly
what types of goods would result in $dQ = 0$? [inelastic!]

so if we care about efficiency, a low DWL, what types of goods should be taxed? [more inelastic]

DWL

These observations lead to two important rules of thumb for reducing the efficiency costs of a tax:

- what you should do if you want to reduce the efficiency costs of a tax
1. The Inverse Elasticity Rule: The more inelastic the demand for a good is, the more it should be taxed
 2. The Base-Broadening Rule: It is better to tax a wide variety of goods at a moderate rate than to tax very few goods at a high rate.

Of course, as rules of thumb, there are many exceptions. Furthermore, these rules do not take into account any distributional or fairness concerns. For instance, suppose the government could only tax caviar or insulin. The demand for caviar is likely more elastic, so the implication of these rules is that you ought to tax insulin at a higher rate than caviar. However, many people would view that system as unjust. In practice, many essential commodities face lower tax rates. For instance, the California state sales tax does not apply to groceries or prescription medication.

equity and efficiency tradeoff!

↳ social welfare functions

2 Practice problems

2.1 Gruber, Ch.20, Q.2

The government of Washlovia wants to impose a tax on clothes dryers. In East Washlovia the demand elasticity for clothes dryers is -2.4 while in West Washlovia the demand elasticity is -1.7. Where will the tax inefficiency be greater for the same level tax? Explain.

$$\frac{\text{east}}{\epsilon_D = -2.4} \quad \bigg| \quad \frac{\text{west}}{\epsilon_D = -1.7}$$

same level of tax!

might be inclined to start here but we don't anything about ϵ_S

$$DWL = \frac{1}{2} dQ dt = \frac{1}{2} \frac{\epsilon_S \cdot \epsilon_D}{\epsilon_S - \epsilon_D} \cdot \frac{Q}{P} (dt)^2$$

how about the inverse elasticity rule!

↳ to reduce efficiency cost, tax more inelastic demand (west is more inelastic!)

↓

BUT, question asks "where will the tax inefficiency be greater?"

↓

more likely to Δ their behavior → more distortion or more elastic side: east! (less efficient)

2.2 Gruber, Ch.20, Q.10 (adapted)

$$DWL = \frac{1}{2} \cdot dQ \cdot dt$$

The market demand for pet turtles is $Q = 2,600 - 20P$, and the government intends to place a \$4 per turtle tax on pet turtle purchases. Calculate the deadweight loss of this tax when:

1. Supply of pet turtles is $Q = 400$.
2. Supply of pet turtles is $Q = 12P$.
3. Explain why the deadweight loss calculations differ between 1 and 2.

① $Q = 400$, perfectly inelastic supply
 $dQ = 0$, 400 before and after tax!
 $DWL = \frac{1}{2} \cdot (0)(4) = 0$

[See math below]

② before tax quantity

$$2600 - 20P = 12P$$

$$2600 = 32P$$

$$P = \$81.25$$

$$Q = 12(81.25) = 975$$

$$\begin{aligned} DWL &= \frac{1}{2} \cdot dQ \cdot dt \\ &= \frac{1}{2} \cdot [975 - 945] \cdot 4 \\ &= 2(30) = 60 \end{aligned}$$

after tax quantity

who bears statutory incidence? not 100% clear
 does it matter? no, will reach same equilibrium
 say producer's pay! (key takeaway from section 3)
 then they keep less of their profit, s.t.

$$2600 - 20P = 12(P - 4)$$

$$2600 - 20P = 12P - 48$$

$$2648 = 32P$$

$$P = \$82.75$$

$$Q = 12(82.75 - 4) = 945$$

say consumer's pay!

$$2600 - 20(P + 4) = 12P$$

$$2600 - 20P - 80 = 12P$$

$$2520 = 32P$$

$$P = \$78.75$$

$$\text{consumer's pay} = 78.75 + 4 = 82.75$$

$$Q = 12(78.75) = 945$$

$$\textcircled{1} Q_s = 400$$

$$2600 - 20P = 400$$

$$2200 = 20P$$

$$P = 110$$

$$Q_1 = 2600 - 20(110) = 2600 - 2200 \\ = 400$$

consumers pay.

$$2600 - 20(P+4) = 400$$

$$2600 - 20P - 80 = 400$$

$$2120 = 20P$$

$$P = 106$$

$$Q_2 = 2600 - 20(106+4)$$

$$= 2600 - 20(110)$$

$$= 2600 - 2200$$

$$= 400$$

$$Q_2 - Q_1 = 400 - 400 = \underline{0}$$